inch of water per minute and the exhaust pipe ... is full an
both pipes are open, how many hours will it take to empty it?

Sol. Volume of the tank = (9 × 5 × 2) cu *ft* = 90 cu. ft.

= (90 × 12 × 12 × 12) cu. inch.

Volume of water drained by the exhaust pipe in one minute

$$= \left(\frac{90 \times 12 \times 12 \times 12}{3 \times 60}\right) \text{ cu. inch} = 864 \text{ cu. inch.}$$

Net volume drained in one minute, when both the pipes are opened = (864 − 576) cu. inch = 288 cu. inch.

$$\therefore \text{ Required time} = \left(\frac{90 \times 12 \times 12 \times 12}{288 \times 60}\right) \text{hrs} = 9 \text{ hrs.}$$

Ex. 2. *Pipe A can fill a tank in 30 hours and pipe B in 45 hours. If both the pipes are opened in an empty tank how much time will they take to fill it?*

Sol. Part filled by A in 1 hour = $\frac{1}{30}$; Part filled by B in 1 hour = $\frac{1}{45}$.

$$\text{Part filled by (A + B) in 1 hour} = \left(\frac{1}{30} + \frac{1}{45}\right) = \frac{10}{180} = \frac{1}{18}.$$

Hence, pipes A and B together will fill the tank in 18 hours.

Ex. 3. *A cistern can be filled by pipes A and B in 4 hours and 6 hours respectively. When full, the cistern can be emptied by pipe C in 8 hours. If all the pipes were turned on at the same time, in how much time will the cistern be filled?*

Sol. Net part filled in 1 hour = $\left(\frac{1}{4} + \frac{1}{6} - \frac{1}{8}\right) = \frac{7}{24}$.

$\therefore$ The cistern will be full in $\frac{24}{7}$ hrs, i.e. $3\frac{3}{7}$ hrs.

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Thus, net part emptied $= \left(\frac{1}{40} - \frac{1}{10}\right) = \frac{1}{40} = \frac{1}{8}$.

Hence, the two pipes together will empty the tank in 8 minutes.

Ex. 5. *A tap can fill a tank in 10 minutes and another can empty it in 6 minutes. If the tank is already two-fifths full and both the taps are opened together, will the tank be filled or emptied? How long will it take before the tank is either filled completely or emptied completely, as the case may be?* (M.A.T., 2004, 05; S.S.C., 2004)

Sol. Clearly, the outlet pipe is faster than the inlet pipe and so, the tank will be emptied.

Part to be emptied $= \frac{2}{5}$.

Net part emptied in 1 minute $= \left(\frac{1}{6} - \frac{1}{10}\right) = \frac{2}{30} = \frac{1}{15}$.

$\therefore \quad \frac{1}{15} : \frac{2}{5} :: 1 : x$ or $x = \left(\frac{2}{5} \times 1 \times 15\right) = 6$ min.

So, the tank will be emptied in 6 minutes.

Ex. 6. *A cistern has two taps which fill it in 12 minutes and 15 minutes respectively. There is also a waste pipe in the cistern. When all the three are opened, the empty cistern is full in 20 minutes. How long will the waste pipe take to empty the full cistern?* (M.A.T., 2005)

Sol. Work done by the waste pipe in 1 minute

$$= \frac{1}{20} - \left(\frac{1}{12} + \frac{1}{15}\right) = -\frac{1}{10}$$ [–ve sign means emptying]

$\therefore$ Waste pipe will empty the full cistern in 10 minutes.

Ex. 7. *An electric pump can fill a tank in 3 hours. Because of a leak in the tank it took $3\frac{1}{2}$ hours to fill the tank. If the tank is full, how much time will the leak take to empty it?*

Sol. Work done by the leak in 1 hour $= \left[\frac{1}{3} - \frac{1}{\left(\frac{7}{2}\right)}\right] = \left(\frac{1}{3} - \frac{2}{7}\right) = \frac{1}{21}$.

$\therefore$ The leak will empty the tank in 21 hours.

Ex. 8. *Two pipes can fill a cistern in 14 hours and 16 hours respectively. The pipes are opened simultaneously and it is found that due to leakage in the bottom it took 32 minutes more to fill the cistern. When the cistern is full, in what time will the leak empty it?* (I.I.F.T., 2005)

Sol. Work done by the two pipes in 1 hour $= \left(\frac{1}{14} + \frac{1}{16}\right) = \frac{15}{112}$.

$\therefore$ Time taken by these pipes to fill the tank $= \frac{112}{15}$ hrs $= 7$ hrs 28 min.

Due to leakage, time taken = 7 hrs 28 min + 32 min = 8 hrs.

$\therefore$ Work done by (two pipes + leak) in 1 hour $= \frac{1}{8}$.

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